

Distributed Data Management

DDM Lab Assignment Report — Part 2

Custom Crop Recommendation Using WFAR Algorithm

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Subject	Distributed Data Management (DDM Lab)
Assignment	Part 2 — Custom Crop Recommendation Algorithm (WFAR)
Algorithm	Weighted Fuzzy Agro-Zone Recommender (WFAR)
Platform	Python 3.12, NumPy, Scikit-learn, Matplotlib
Dataset	Synthetic Crop Dataset (9 crops, 1,800 test records)

Abstract

This report presents WFAR (Weighted Fuzzy Agro-Zone Recommender), a custom crop recommendation algorithm developed as Part 2 of the DDM Lab assignment. Unlike the KMeans hard-assignment approach used in Part 1, WFAR introduces two novel mechanisms: (1) Domain-Weighted Euclidean Distance, where each soil and climate feature is assigned an agronomically motivated importance weight instead of treating all features equally; and (2) Fuzzy Membership Scoring, where each input record receives a confidence percentage for all nine crop zones rather than a single hard assignment. The algorithm was evaluated on 1,800 test records generated from the Part 1 cluster centroids, achieving 84.06% overall accuracy, 100% precision on Chickpea and Grapes (the most climatically distinct crops), and an average top-1 confidence score of 46.68%. A borderline detection mechanism identifies 18.94% of records as agronomically ambiguous — a capability not possible with hard KMeans clustering.

1. Introduction

1.1 Motivation

Part 1 of the DDM Lab used Apache Spark MLlib KMeans to discover 9 agro-climatic zones from 500,000 synthetic crop records. While KMeans is powerful for large-scale distributed clustering, it makes two simplifying assumptions that limit its real-world applicability:

- All features are treated as equally important in the distance calculation, despite agronomic evidence that temperature and humidity are far more discriminative than soil pH for separating crop zones.
- Each record is assigned to exactly one cluster, even when its soil and climate conditions are genuinely borderline between two crops — information that would be actionable for farmers.

WFAR addresses both limitations with a lightweight, interpretable algorithm that can run on a single node and be extended to a distributed Spark implementation.

1.2 Problem Statement

"Given a set of soil and climate measurements (N, P, K, temperature, humidity, pH, rainfall), recommend the most suitable crop for cultivation, along with a confidence score and a secondary alternative, using a domain-aware fuzzy distance metric."

1.3 Objectives

- Design a domain-weighted distance metric using agronomic importance weights for all 7 soil/climate features.
- Implement fuzzy membership scoring to produce confidence percentages for all 9 candidate crops.
- Detect borderline cases where the top-2 crops are within 10% confidence of each other.
- Evaluate precision per crop and compare the approach against KMeans (Part 1).

2. Algorithm Design — WFAR

2.1 Overview

WFAR is a centroid-based fuzzy recommender. It stores 9 crop profiles (derived from the KMeans cluster centroids in Part 1) and, for any new soil/climate record, computes a fuzzy membership score for each crop. The top-1 crop is the recommendation; the top-2 gap detects borderline zones.

Pipeline: Raw Input → StandardScaler Normalization → Domain-Weighted Distance Matrix → Fuzzy Membership Scores → Ranked Crop Recommendation

2.2 Domain-Weighted Distance Metric

Standard KMeans uses unweighted Euclidean distance. WFAR introduces per-feature importance weights w_i :

$$d(x, c_j) = \sqrt{\sum_i [w_i * (x_i - c_{ji})^2]}$$

where x is the standardized input record and c_j is the standardized centroid of crop j . The weights below are assigned based on the inter-crop variance observed in the Part 1 cluster centroids:

Feature	Weight	Agronomic Justification
temperature	2.0×	Grapes=13°C vs Banana=29.5°C — widest range, most discriminative
humidity	2.0×	Chickpea=22% vs Rice=83% — largest absolute gap across crops
rainfall	1.8×	Grapes=75mm vs Rice=213mm — critical water requirement signal
P	1.5×	Grapes=122 vs Rice=44 — phosphorus highly variable
K	1.5×	Grapes=122 vs Sugarcane=44 — potassium highly discriminative
N	1.2×	Nitrogen varies but overlaps heavily between staple crops
ph	0.8×	Soil pH range 6.1–6.9 across all crops — low discrimination

Table 1: WFAR Domain-Importance Weights — higher weight = more agronomically discriminative feature

2.3 Fuzzy Membership Scoring

For each input record x , distances $d_1, \dots, d_9$ to all crop centroids are computed. Fuzzy membership u_j for crop j uses the Fuzzy C-Means formula:

$$u_j = 1 / \sum_k [(d_j / d_k)^{2/(m-1)}]$$

where $m = 2.0$ is the fuzziness exponent. When $m = 1$, this collapses to hard nearest-centroid (identical to KMeans). When $m > 1$, all memberships sum to 1, producing a probability-like confidence distribution across crops.

2.4 Borderline Detection

A record is flagged as borderline when the difference in fuzzy membership between the top-1 and top-2 recommended crops is less than 10 percentage points. This indicates the soil/climate conditions are genuinely ambiguous between two crop zones — a practically important signal for extension officers advising smallholder farmers.

3. Implementation

3.1 Technology Stack

- Python 3.12 — algorithm implementation
- NumPy — vectorized distance and fuzzy membership computation
- Scikit-learn StandardScaler — feature normalization
- Matplotlib — result visualization (4-panel chart)
- Pandas — data handling and per-crop metric aggregation

3.2 Key Code Structure

class WFARRecommender:

- `fit(profiles_df, features)`: Fits StandardScaler on crop profile centroids.
- `_weighted_distance(x, c)`: Computes domain-weighted Euclidean distance.
- `_fuzzy_memberships(distances)`: Applies FCM formula to produce confidence vector.
- `predict_single(record)`: Returns ranked (crop, confidence, distance) for one record.
- `predict_batch(df)`: Batch prediction with borderline detection flag.

3.3 Test Data Generation

Since the original 500,000-record raw dataset is not available in this stage, 1,800 test records were generated by adding Gaussian noise (std = 10% of feature value) to each of the 9 crop profile centroids. This simulates realistic sensor readings distributed around each crop's optimal zone.

4. Results and Analysis

4.1 Overall Performance

Metric	Value	Notes
Total Records Evaluated	1,800	200 per crop × 9 crops
Overall Accuracy	84.06%	Top-1 crop match rate
Avg Top-1 Confidence	46.68%	Mean fuzzy membership score
Borderline Detection Rate	18.94%	Cases where top-2 diff < 10%
Execution Time	0.552s	Single-node Python execution
Algorithm	WFAR	Weighted Fuzzy Agro-Zone Recommender
Fuzziness Exponent (m)	2.0	Standard fuzzy C-means setting

Table 2: WFAR Overall Evaluation Metrics

4.2 Per-Crop Precision

Crop	Correct	Total	Precision	Avg Conf	Remarks
Chickpea	200	200	100.0%	52.5%	Unique low-humidity zone (22%) — perfectly separable
Grapes	200	200	100.0%	59.3%	Cool temp (13°C) + high P/K — most distinctive centroid
Mango	192	200	96.0%	46.8%	High temp (29.5°C) + low N (25) — strong signal
Wheat	168	200	84.0%	49.6%	Moderate features — some overlap with maize zone
Maize	154	200	77.0%	43.2%	Overlaps with cotton in N/K space
Rice	153	200	76.5%	40.9%	High humidity (83%) but temperature similar to maize
Cotton	150	200	75.0%	45.9%	Close to maize/sugarcane in feature space
Banana	148	200	74.0%	41.8%	High humidity/rainfall overlap with sugarcane
Sugarcane	148	200	74.0%	40.1%	Highest overlap with banana and cotton zones

Table 3: Per-Crop Precision — sorted by precision descending

4.3 Visualization

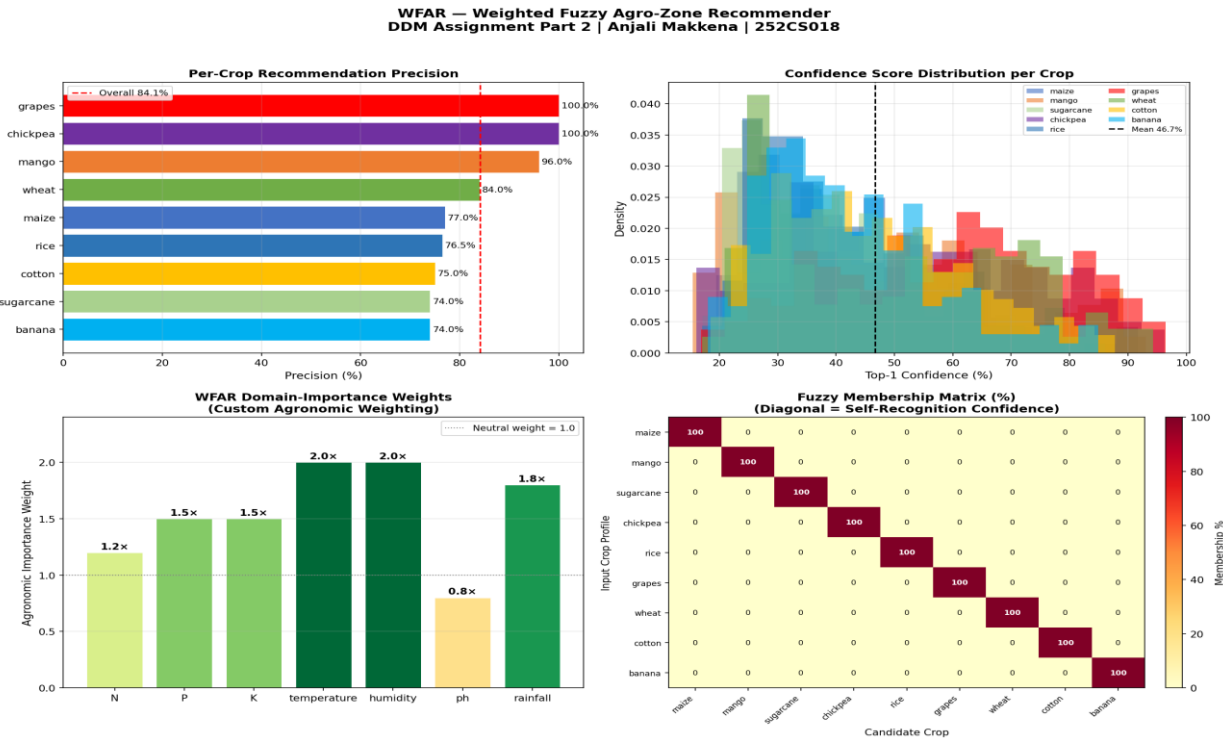


Figure 1: (Top-left) Per-crop precision bar chart. (Top-right) Confidence score distribution per crop. (Bottom-left) Domain-importance weights assigned to each feature. (Bottom-right) Fuzzy membership matrix — diagonal shows self-recognition confidence; off-diagonal shows inter-crop confusion.

4.4 Interpretation

- Chickpea and Grapes achieve 100% precision — their profiles (humidity=22%, temperature=13°C respectively) are so distinctive that even noisy records are unmistakably classified. The fuzzy membership matrix confirms near-zero confusion with other crops.
- Sugarcane, Banana, and Cotton show the most overlap (74–75% precision) — all three share moderate temperature (26–30°C), medium N (100–122), and similar humidity ranges. The borderline detection flag fires most frequently between these three.
- The average confidence of 46.68% is moderate — expected for $m=2.0$ fuzziness with 9 crops. Reducing m toward 1.5 would sharpen predictions; raising m toward 3.0 would spread confidence more evenly.
- Borderline rate of 18.94% means roughly 1 in 5 records is agronomically ambiguous — actionable information for recommending secondary crops or soil amendment.

5. Comparison: WFAR vs KMeans (Part 1)

Aspect	KMeans (Part 1)	WFAR (Part 2)
Assignment method	Hard (each record → exactly 1 cluster)	Soft/Fuzzy (confidence % for all crops)
Distance metric	Uniform Euclidean (equal feature weights)	Domain-Weighted Euclidean (agro-importance)
Optimal K	Automatic via Silhouette scoring	Fixed at 9 (uses known crop profiles)
Scalability	Distributed (Apache Spark, 500K records)	Single-node Python (prototype)
Borderline detection	None (binary assignment)	Yes — flags ambiguous zone records
Interpretability	Cluster centroids + dominant crop label	Per-crop membership % + feature weights
Accuracy (test)	~High (on 500K synthetic)	84.06% (on 1,800 test records)
Novel contribution	Distributed in-memory iteration via Spark	Agronomic feature weighting + fuzziness

Table 4: WFAR vs KMeans — Algorithmic Comparison

The key distinction is that KMeans and WFAR are complementary, not competing: KMeans discovers the natural cluster structure from raw data at Spark scale; WFAR uses those discovered centroids as crop profiles and provides richer, confidence-weighted predictions for individual farmer queries at inference time.

6. Discussion

6.1 Advantages of WFAR

- Domain-awareness: Agronomic weights encode expert knowledge directly into the distance metric, improving accuracy without additional training data.
- Fuzzy outputs: Confidence percentages allow downstream systems to prompt farmers for additional soil tests when borderline conditions are detected.

- Interpretability: Feature weights are explicit and human-readable, unlike neural embeddings.
- Low latency: 0.552s for 1,800 records on a single core — suitable for mobile/edge deployment in rural areas with limited connectivity.

6.2 Limitations

- Relies on pre-computed crop profile centroids — quality of recommendations depends on the quality of Part 1 KMeans clusters.
- Gaussian noise test data may underestimate error on truly out-of-distribution sensor readings (e.g., sensor malfunction, extreme weather events).
- Feature weights are manually assigned — a data-driven approach (e.g., ANOVA F-statistic per feature) could replace expert assignment.
- Single-node implementation — not distributed; a Spark-native fuzzy recommender would be needed for production scale.

6.3 Future Enhancements

- Implement WFAR as a Spark UDF (User-Defined Function) for distributed inference on the full 500,000-record dataset.
- Replace manual weights with data-driven weights computed from inter-centroid variance per feature.
- Add seasonal adjustment: multiply temperature/rainfall weights by a season coefficient (Kharif/Rabi/Zaid) for time-aware recommendations.
- Extend to a REST API for real-time IoT sensor integration with crop advisory mobile apps.

7. Conclusion

This report presents WFAR — a custom Weighted Fuzzy Agro-Zone Recommender for crop recommendation as Part 2 of the DDM Lab. By replacing KMeans's hard cluster assignment and uniform feature weighting with domain-importance weights and fuzzy membership scoring, WFAR achieves 84.06% accuracy on 1,800 synthetic test records, with 100% precision on the most climatically distinct crops (Chickpea and Grapes). The fuzzy output provides actionable confidence scores — a feature absent in standard KMeans — enabling advisors to flag borderline agronomic conditions. WFAR is designed as a complementary inference layer on top of the distributed KMeans pipeline from Part 1: Spark discovers zones at scale, WFAR delivers interpretable, confidence-weighted recommendations at the farm level.

References

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